

Jarvis Market Data Service - API Specification

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This document specifies the HTTP API contract for the FastAPI-based market data layer used by the Jarvis AI investment analysis pipeline. It is written for developers integrating n8n workflows, Supabase triggers, and downstream analytics.

1. Scope and responsibilities

Jarvis Market Data Service is responsible for: (a) fetching market time-series data (OHLCV) and basic ticker fundamentals, (b) normalising outputs into JSON that is easy to persist in Supabase, and (c) resolving ticker symbols to legal company names for patent matching.

2. Base URL, transport, and content type

All endpoints are served over HTTPS and accept/return JSON. Requests must include

Content-Type: application/json.

Environments: In MVP, this service is typically deployed as a Docker container on an EC2 instance (or equivalent). Expose it behind a reverse proxy with TLS (recommended) and restrict access to trusted callers (n8n, internal tools).

3. Authentication and authorisation

MVP deployments commonly run in a trusted network segment and do not require per-request auth. For production, add one of: (a) an API key header, (b) mTLS, or (c) allow-listing by IP + reverse proxy auth. Treat this service as privileged because it can trigger paid API usage and downstream workflows.

4. Common conventions

Dates use ISO-8601 (YYYY-MM-DD). **Interval** uses vendor-compatible strings such as 1d, 1h. **batch_id** is a caller-provided correlation identifier; in the Jarvis pipeline it is set equal to the per-ticker **request_id**.

5. Endpoint: POST /portfolio/json

Fetches OHLCV time-series and basic ticker metadata. While the request supports multiple tickers, the recommended usage is **one ticker per call** to enable parallelism in n8n and to minimise the blast-radius of rate limiting or data-quality failures.

Request body schema

Field	Type	Required	Description
tickers	string[]	Yes	List of ticker symbols. Recommended length = 1 in MVP.

start_date	string	Yes	Start of analysis period (YYYY-MM-DD).
end_date	string	Yes	End of analysis period (YYYY-MM-DD).
interval	string	No	Candle interval. Default: 1d.
batch_id	string	No	Correlation id echoed in response. In Jarvis: set to request_id.
fetch_mode	string	No	Output profile; e.g. chart. Default: chart.
strict	boolean	No	If true (default), missing/invalid data returns an error. If false, returns

Response schema (200 OK)

Field	Type	Required	Description
(root)	object[]	Yes	Array of per-ticker results (one element per ticker).
ticker	string	Yes	Ticker symbol.
batch_id	string	No	Echoed correlation id (if provided).
raw_ohlcv	object[]	Yes	Array of candles: {datetime, open, high, low, close, volume}.
raw_tickerinfo	object	Yes	Fundamentals/metadata: sector, marketCap, beta, forwardPE, etc (fi
error	object	No	Present when strict=false and a ticker fails. Contains {code, message

Example request (one ticker)

```
curl -X POST https:///portfolio/json \ -H "Content-Type: application/json" \ -d '{
  "tickers": ["AAPL"], "start_date": "2023-01-01", "end_date": "2024-01-01",
  "interval": "1d", "batch_id": "45b1d1d4-66a9-4a03-864f-326da73adfe", "fetch_mode":
  "chart", "strict": true }'
```

Example response (shape only)

```
[ { "ticker": "AAPL", "batch_id": "45b1d1d4-66a9-4a03-864f-326da73adfe",
  "raw_ohlcv": [{ "datetime": "2023-01-03", "open": 130.28, "high": 130.90, "low": 124.17, "c
  lose": 125.07, "volume": 112117500}, "..."], "raw_tickerinfo":
  { "sector": "Technology", "marketCap": 2500000000000, "beta": 1.24, "forwardPE": 28.1,
  "..."} } ]
```

6. Endpoint: POST /company/name

Resolves a ticker symbol to the company’s legal name, suitable for downstream patent matching (PatentsView expects assignee names, not tickers).

Request body schema

Field	Type	Required	Description
ticker	string	Yes	Ticker symbol (e.g., AAPL, SAP.DE).

Response schema (200 OK)

Field	Type	Required	Description
company_name	string	Yes	Resolved legal name (e.g., Apple Inc.).
ticker	string	No	Echoed ticker (if returned by implementation).
source	string	No	Resolution source (e.g., yahoo_search).

Example request/response

```
curl -X POST https:///company/name \ -H "Content-Type: application/json" \ -d
'{"ticker": "AAPL"}'

{"company_name": "Apple Inc."}
```

7. Error handling

Scenario	HTTP	Recommended caller action
Rate limited by upstream (HTTP 429)	429	Throttle (2-3s per ticker), retry with backoff; consider proxy rotation / paid o
Upstream returns empty payload / IP blocked	502 or 424	Wait and retry later; rotate IP/User-Agent; alert operator.
Invalid ticker / no data	400	Mark track_requests as failed; continue remaining tickers.
Timeout talking to upstream	504	Retry with exponential backoff (3s → 6s → 12s).

8. Performance and operational notes

Expected throughput (MVP): constrained primarily by upstream rate limits, not by FastAPI compute. A safe baseline is 1 request per ticker every 2-3 seconds per worker. Scale horizontally by running multiple workers if upstream policies allow it.

Observability: log request start/end with batch_id, ticker, duration, and upstream status. This is essential to map failures back to track_requests and to support retries.

Appendix: Field notes for Jarvis integrators

In Jarvis, n8n calls POST /portfolio/json once per ticker and writes the response to ticker_data.raw_ohlcv (JSONB). Use the same request_id as batch_id so that all logs, rows, and artefacts correlate cleanly across services.

End of document.